Deprivation of Parenting Disrupts Development of Homeostatic and Reward Systems in Marmoset Monkey Offspring

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Background: *Early environment is a major determinant of long-term mental health, evidenced by the relationship between early-life neglect or abuse and chronically increased vulnerability to developmental psychopathology, including major depressive disorder (MDD). Animal studies can increase understanding of environmentally mediated causal risk processes. We describe how daily deprivation of biological parenting in primate infants disrupts development of homeostatic and reward systems central to MDD.*

Methods: *Nine breeding pairs of marmoset monkeys provided control twins (CON) and early-deprived twins (ED); the latter were socially isolated for 30–120 min/day on days 2–28. During the first year of life, basal urinary norepinephrine (NE) titers and cardiophysiologic activity were measured. At the end of year 1 (adolescence), automated neuropsychologic tests were conducted to measure responsiveness to changes in stimulus-reward association (simple/reversed visual discrimination learning) and to reward per se (progressive ratio [PR] reinforcement schedule).*

Results: *The ED monkeys exhibited increased basal urinary NE titers and increased systolic blood pressure relative to CON siblings. The ED monkeys required more sessions to reinstate stimulus-oriented behavior following reversal, suggesting increased vulnerability to perceived loss of environmental control; ED monkeys also performed less PR operant responses, indicating that reward was less of an incentive and that they were mildly anhedonic relative to CON.*

Conclusions: *In marmoset monkeys, neglect-like manipulation of ED leads to chronic changes in homeostatic systems, similar to those in children and adolescents exposed to early-life adversity and in MDD, and to responses to environmental stimuli similar to those that characterize MDD.*

Key Words: Allostatic load, depression, developmental psychopathology, early deprivation, nonhuman primate, vulnerability traits

Adverse events occurring early in mammalian life can exert acute and long-term influences on the brain and on physiology and behavior. Parenting constitutes the major component of the infant's environment, in the form of predictable multisensory stimulation that nurtures offspring homeostasis and development. Parent–infant behavior is inadequate when it fails to promote homeostasis and renders the infant vulnerable to external stressors. In humans, the clinical and epidemiologic evidence indicates that early-life stress in the form of parental loss, neglect, or abuse leads, in some individuals, to altered physiologic and neurobiological status and increases the likelihood that they will suffer developmental and adulthood affective psychopathology, including posttraumatic stress disorder (PTSD) and major depressive disorder (MDD) (Caspi et al 2003; Heim and Nemeroff 2001; Kendler et al 2002; Penza et al 2003; Rutter 2003; Teicher et al 2003). As is the case for adulthood-onset affective disorders, physiologic and neurobiological abnormalities are frequent in children who have suffered maltreatment and are presenting with affective disorder. There is also the possibility that even in the absence of clinical symptoms, abnormal traits are present that mediate the high long-term vulnerability exhibited by sufferers of early-life adversity. Thus, children with affective disorder secondary to past maltreatment demonstrate elevated urinary catecholamine titers (De Bellis et al 1999). Maltreated children with PTSD exhibit increased systolic blood pressure and heart rate (Perry 1994). Studies investigating hypothalamic-pituitary-adrenal (HPA) function in maltreated children have not yielded consistent findings, but there is evidence that severely neglected children with psychopathology exhibit low peak basal glucocorticoid (GC) levels (De Bellis et al 1999). Furthermore, young women who were abused as children also exhibit lower basal GC than control women, and this is the case for young women both with and without current MDD (Heim et al 2001).

In animals, the importance of the maternal environment to infant homeostasis and long-term offspring development has been studied extensively in laboratory rats. Separation of the litter from the mother for a single period of 8 hours or repeated isolation of pups for 1 hour/day leads to acute increases in adrenocorticotropic hormone (ACTH) and GC (Levine 1994; McCormick et al 1998). Although these HPA hormones have received the most experimental attention, neuronal and glial development, bioamine neurotransmitter systems, anterior pituitary and adrenomedullary hormones, and brown adipose tissue thermogenesis provide further examples of rat pup systems that respond acutely to such maternal deprivation (Blumberg and Sokoloff 1998; Hofer 1994a, 1994b; Zhang et al 2002). With regard to chronic effects, repeated separation of the litter from the mother for 3 hours/day yields adult offspring that exhibit increased HPA responsiveness and increased behavioral anxiety to psychologic stressors (Huot et al 2001; Ladd et al 2000; Liu et al 2000; Plotsky and Meaney 1993; Pryce and Feldon 2003). Chronic effects analogous to the symptoms and associated changes of affective disorders have received relatively little attention to date: repeated maternal separation did not lead to reduced appetitive interest in reward (i.e., anhedonia, a major symptom of MDD; Shalev and Kafkafi 2002), although it did lead to increased brain norepinephrine (NE; Liu et al 2000), in common with human MDD (Wong et al 2000). Also, long-term anxiogenic effects of repeated maternal separation are attenu-

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ated by monoamine-targeting antidepressant treatment (Huot et al 2001).

The first animal model studies of depression were in fact those that used maternal separation of young primate infants to increase understanding of the anaclitic depression and protest–despair–detachment syndromes described in human infants exposed to maternal deprivation (Bowlby 1962; Spitz and Wolf 1946). In mature (6-month-old) rhesus macaque infants, maternal deprivation for 2–3 weeks induces protest and despair (Hinde et al 1966; Seay et al 1962; Spencer-Booth and Hinde 1971a, 1971b). Although there was no evidence of detachment at reunion in these studies, marked long-term effects were observed in 1 year olds (juveniles), including increased proximity to the mother, reduced locomotor activity, and increased anxiety (Spencer-Booth and Hinde 1971b). Also in rhesus macaques, there are definitive studies of the long-term effects on neurobiology, physiology, and behavior of maternal privation, that is, infant isolation at birth, such that it develops in the absence of the mother (Harlow 1958; Harlow and Harlow 1962; Kraemer 1992). Maternally deprived macaques develop into individuals with reduced basal NE in cerebrospinal fluid, disturbed social behavior including impulsive and hyperaggressive interactions, and reduced interest in palatable rewards (Kraemer et al 1989; Paul et al 2000). Genotype is important in determining the maternal privation outcome: recent studies have demonstrated that the long-term effects of privation on serotonin activity and behavior are dependent on the serotonin transporter genotype (Bennett et al 2002; Champoux et al 2002). Integrating the findings in primate maternal privation research, the psychobiological attachment theory (Kraemer 1992) proposes that the absence of the mother causes abnormal physiology and behavior precisely because it causes abnormal brain development.

Although maternal privation certainly occurs in human societies (e.g., O'Connor et al 2003), its pathologic effects on human development could well be fundamentally different from those of deprivation events, such as physical and emotional neglect, that are embedded within extant parent–infant relationships (Rutter 2003). In this latter context, there is a great need for primate studies of the impact of neglect-like, repeated maternal deprivation on those neurobiological, physiologic, and behavioral systems that are of direct relevance to human early-life adversity and the developmental psychopathologies associated with it. Such an approach could potentially yield environmental models of high validity with respect to the systems affected, the mediating causal processes, and therapeutic responsiveness. In the common marmoset monkey (*Callithrix jacchus*), we recently described the effects of this type of manipulation on anthropometry, endocrinology, and behavior across infancy (Dettling et al 2002b). This is a small-bodied simian primate characterized by twin births and intensive biparental care (Pryce 1996). The early-deprivation (ED) procedure causes acute, daily increases in urinary GC, epinephrine, and NE (when immediate post-ED and pre-ED titers are compared) and yields infants of lower body weight and with lower basal urinary GC than control animals (the nonmanipulated siblings of ED marmosets). In the home cage, ED and CON infants receive similar amounts of parental care, but the former emit more distress vocalizations and engage in less social play (Dettling et al 2002b).

Here we report on the status of these ED marmoset monkeys across the first year of life and describe how ED leads to abnormal functioning of homeostatic and reward systems as assessed using physiologic and psychologic parameters, respectively. The rationale for focusing on homeostatic systems is the evidence, reviewed earlier, for chronically altered functioning of homeostatic endocrine and cardiophysiologic systems in children with a history of early-life adversity and in individuals, including such children, with affective disorder, including MDD. Basal urinary NE was determined in first-void urine samples that could be collected using a stress-free procedure from age 3 months (late infancy–early juvenile phase) onward. Basal blood pressure and heart rate were determined for continuous 24-hour periods in juveniles using surgically implanted radiotransmitters and telemetry. The rationale for focusing on reward systems is the evidence for altered responsiveness to environmental reinforcers in MDD. Using a visual discrimination learning and reversal task, we wanted to investigate the effects of ED on perseveration, which is increased in depression (Silberman et al 1983), and on responsiveness to (perceived) loss of environmental control, relating to the cognitions of loss of coping ability and control that are central to MDD (American Psychiatric Association [APA] 1994; Meerum Terwogt and Stegge 2001). Using a progressive ratio reinforcement schedule, we wanted to investigate the effects of ED on motivation to obtain reward, with direct relevance to the defining MDD symptom of anhedonia (APA 1994). Subjects could be trained on these tasks from late juvenility and tested at the adolescent stage (age 9–12 months).

Methods and Materials

Subjects and Early Deprivation

All procedures were performed under permit in accordance with the Swiss Animal Protection Act of 1978. Nine established breeding pairs of common marmosets living in cages of 3–4 m^3 each contributed two sets of twin offspring to this study, with an interbirth interval of 12 months (Table 1). In the case of four pairs, the first set of twins was allocated to the early-deprivation (ED) treatment and the second set to the control (CON) treatment; and in the other five pairs, the reverse was the case. In total, there were 11 CON male, 7 CON female, 9 ED male, and 9 ED female subjects. The first set of twins was euthanized at age 11–12 months, and the second set of twins was born soon thereafter, such that all subjects developed in a family group that included the twin sibling and the biological parents only (Dettling et al 2002b). On each of postnatal days 2–28 (weeks 1–4), ED was conducted for between 30 and 120 min/day, beginning at different times from 8:30 AM to 5:00 PM, for a total of 9 hours/week. Duration and time of day of ED were varied to try and increase stress impact in relation to unpredictability. Early deprivation comprised restraining the carrying parent in the home cage, removing the infant, and then transferring the infant to a remote isolation chamber. Early deprivation was conducted successively with each infant in a pair, so that one infant remained with the parents at all times. The control procedure consisted of the carrying parent being restrained briefly and then released to simulate this component of the ED procedure (for further details, see Dettling et al 2002b).

Immunoreactive Estimation of Basal Urinary Norepinephrine Titers

In the case of the ED and CON twins from five breeding pairs (overall $n = 20$, 7 CON male, 3 CON female, 6 ED male, 4 ED female subjects; Table 1), from age 9–12 weeks (month 3) until sacrifice at age 48–52 weeks (month 12), a first-void urine sample was collected once per week by entering the home cage at 8:00 AM and clean-catching the urine in a plastic container held underneath the target animal, followed by storage at −80°C

Table 1. Summary of Study Parents and Subjects and of Subject Allocation to the Various Parameters Measured

Mother–Father Pair	Subjects[a]	Urinary NE	Cardiophysiology	VD/VR/VRR	PR Schedule
1	CON (F, M); ED (F, M)	√		√	√
2	ED (F, M); CON (F, M)	√		√	√
3	CON (F, M); ED (F, M)	√		√	√
4	ED (F, F); CON (F, M)				√
5	CON (M, M); ED (M, M)	√		√	√
6	ED (F, M); CON (M, M)	√		√	√
7	ED (F, F); CON (F, F)		√		√
8	CON (M, M); ED (F, M)		√		
9	CON (F, M); ED (M, M)		√		

CON, control; ED, early deprivation; F, female; M, male; NE, norepinephrine; VD/VR/VRR, discrimination learning task based on visual discrimination, reversal, re-reversal; PR, progressive ratio schedule of reinforcement task. √ denotes that a parameter was measured in all four subjects from the corresponding breeding pair.

[a]The order in which CON and ED are given denotes the order in which litters were allocated to these two treatments within each breeding pair

before analysis. Pooled monthly samples were prepared for age 3 to 12 months for each subject, and determination of NE titers was performed using a commercial radioimmunoassay (Labor Diagnostika Nord, Nordhorn, Germany) validated for use in the marmoset (Dettling et al 2002b). Using the coefficient of variation, interassay precision was estimated as 11.7% (n = 15). Norepinephrine titers were expressed relative to urinary creatinine (CR) titers (ng NE/mg CR) to control for variation in urine volume/concentration, with CR measured using a colorimetric assay and with an interassay precision of 2.3% (n = 15).

Telemetric Measurement of Basal Cardiophysiology

In the case of the ED and the CON twins from three breeding pairs (overall n = 12; 3 CON male, 3 CON female, 3 ED male, 3 ED female subjects; Table 1), at age 15 weeks, animals were surgically implanted with a telemetric pressure radiotransmitter for the measurement of blood pressure (BP; mm Hg) and heart rate (beats/min). The transmitters used weighed 4 g (model TA11PA-C20, Data Sciences International, St. Paul, Minnesota), and the radio signal was detected up to a distance of 30 cm by six biotelemetry receivers distributed on the ceiling of the home cage and one receiver positioned beneath the sleeping box. The surgical procedure and further details of the methods used for data capture and analysis were as described in Schnell and Wood (1993). One CON male subject had to be euthanized due to postsurgery lameness of the hind limbs, so that five CON and six ED subjects were included in this study. At age 18 weeks, each subject was recorded continuously for 3 × 24 hours, with an average value for systolic BP, diastolic BP, and heart rate calculated for each 1-min block; these data were collapsed into hourly averages per subject and then further collapsed into average daytime and nighttime values per subject.

Home Cage Neuropsychologic Testing of Stimulus-Reward Learning and Motivation

Twins from seven breeding pairs (overall n = 28, 8 CON male, 6 CON female, 7 ED male, 7 ED female subjects; Table 1) were trained and tested on neuropsychologic tasks that are part of the Cambridge Neuropsychologic Test Automated Battery (CANTAB; Roberts et al 1988; Spinelli et al 2004). Using a Monkey CANTAB Testing Station (Cambridge Cognition, Cambridge, United Kingdom) subjects were reinforced by making correct responses to visual stimuli presented on a touch-sensitive computer screen. Training based on classical and operant conditioning stages was conducted following the procedures described in Pearce et al (1998), Roberts et al (1988) and Spinelli et al (2004). Subjects were only 7–8 months of age when training began, and therefore it was important to maintain them with their family group. To enable this, all behavioral training and testing was conducted in the home cage, made possible by enclosing the subject in an inner cage (70 × 50 × 55 cm) suspended on the inside surface of the home cage front. A pulley door to the inner cage could be operated by the experimenter from outside the home cage. Once the target monkey was enclosed in the inner cage, it had exclusive access to a test box that replaced the sleeping box during periods of training and testing (for further details, see Spinelli et al 2004). Reinforcement was in the form of 6 sec of banana-flavored milk drink delivered via a peristaltic pump at a rate of .02 mL/sec to a licker positioned in the middle of the touch-sensitive screen. Using 4–5 training sessions per week, each of either 60-trial maximum or 15-min duration, all subjects could be trained within 3–5 weeks to perform trials comprising a single touch of a single blue square stimulus presented at random to the left or right of the licker that resulted in delivery of reward.

Subjects were then trained to perform simple visual discrimination learning. Two blue visual stimuli that differed in shape were presented on the touch screen; responding to one shape (S+) resulted in reward and to the other (S−) resulted in punishment timeout (Roberts et al 1988; Spinelli et al 2004). Visual discrimination learning (VD) was defined as eight consecutive S+ responses, and subjects were trained to this criterion on three novel stimulus pairs. All four subjects in each of five groups (7 CON male, 3 CON female, 6 ED male, 4 ED female subjects; Table 1) completed training; in the two remaining groups, one or both of the ED twins did not complete VD training, and these groups were dropped from the visual discrimination study. At age 9 months, testing began with subjects being presented with a fourth novel stimulus pair and VD was measured as the number of errors to criterion. Then the stimulus-reward association was reversed such that monkeys were now reinforced for responding to S− (visual reversal [VR] learning); VR was followed by re-reversal to the original association (VRR learning). Completion of one stage to criterion was followed by immediate switching to the next stage, within session. At each stage (VD, VR, VRR), the average number of trials per session, average percentage S+ responses per session, and the cumulative number of errors to reach criterion, were calculated. For VR and VRR, the effects of ED on perseverative responding and chance-level responding were of particular interest. A perseverative session was defined as one in which the percentage of responses to the former S+

was so high that its probability was less than .05 according to a binomial probability distribution with $p = q = .5$ and the number of trials (n) = 50, that is, the average number of trials that subjects performed per session. According to this definition, perseveration occurred in VR/VRR sessions if more than 65% of responses were to the former S+. Reversal was defined as more than 65% of responses to the new S+ (former S−). Between perseveration and reversal were "chance-level" sessions; our prior studies of reversal revealed that during chance-level sessions, marmosets usually responded preferentially to one side of the CANTAB screen (left or right), regardless of stimulus identity, before they began to "follow" and preferentially respond to the new S+. Therefore, the three classical stages of reversal learning (Mackintosh 1974; Sutherland and Mackintosh 1971) were of interest in this study.

After completion of the VD/VR/VRR task, subjects were not tested for 1 week, and then these subjects and the subjects from the two groups that did not complete the VD training were tested on the CANTAB progressive ratio (PR) schedule of reinforcement (Spinelli et al 2004). The PR schedule is a test of the subject's motivation to obtain innate reward in the absence of stimulus–reward discrimination and can be assumed to quantify the incentive to obtain, or the hedonic value of, the reinforcer. To obtain successive reinforcers, marmosets had to perform successively more responses to a large single stimulus on the touch screen; the number of responses required to obtain successive reinforcements increased according to the schedule: 1, 2, 3, 4, 5, 6, 7, 8; 10, 12, . . . 24; 28, 32, and so on. The maximum session duration was 15 min, and the break point (resulting in session termination) was achieved if the subject did not respond to the stimulus at least once within any 5-min period. Following 1 or 2 trial sessions, each subject, now aged 11 months, was given six PR schedule sessions across a 10-day period, and the average total number of responses, total number of reinforcements, final ratio achieved, and time elapsed between session onset and the final touch-screen response were calculated.

Data Analysis

For all parameters studied, a first stage multiple analysis of variance (MANOVA) was performed with between-subject factors of treatment and gender. There were no significant ($p < .05$) main or interaction effects of gender ($p \geq .25$), and this factor was excluded from subsequent analyses. MANOVAs were performed using a randomized-block design with replication, with nine parental breeding pairs each contributing two subjects to two treatments (ED, CON), allowing for analysis of main effects of treatment and parentage and treatment by parentage interaction. Depending on parameter, the MANOVA also included the within-subject factor of age (for urinary NE titers), phase of day (cardiophysiology), and discrimination stage (VD, VR, VRR).

Results

Urinary Basal Norepinephrine Titers

Basal NE was measured in urine samples obtained weekly from CON and ED monkeys aged 3–12 months. An overall decrease in NE titers across this period of development was supported by the significant main effect of the within-subject factor of age [$F(9,90) = 15.73$, $p < .0001$; Figure 1A]. Early-deprived marmosets exhibited abnormally high basal urinary NE titers across infancy to adolescence relative to their CON siblings, as reflected by a significant treatment main effect [$F(1,8) = 19.20$, $p < .02$; Figure 1A]. There was no significant effect involving parentage. Creatinine titers were independent of parentage and treatment ($p \geq .6$).

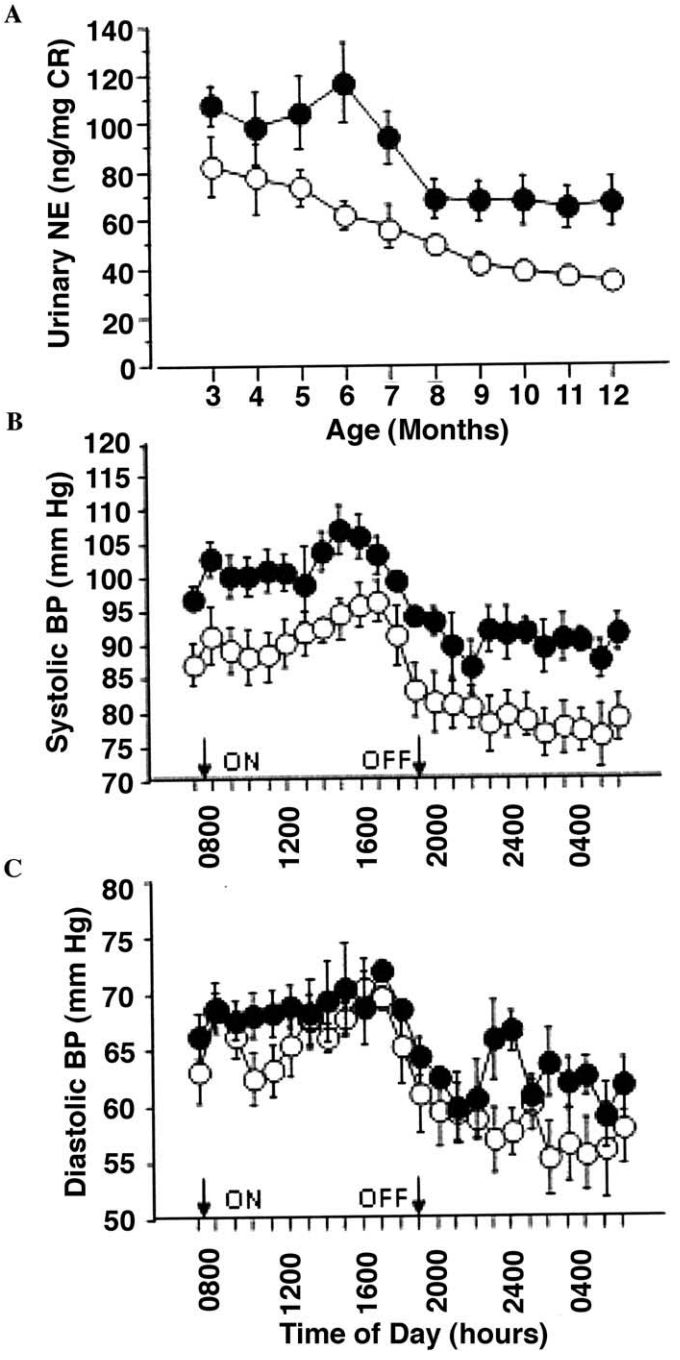


Figure 1. Chronic effects of marmoset early deprivation (filled circles) on autonomic nervous system function relative to control siblings (open circles). **(A)** Mean ± SEM basal urinary noradrenaline titers in morning urine expressed relative to creatinine (ng/mg). **(B)** Mean ± SEM basal systolic blood pressure and **(C)** mean ± SEM basal diastolic blood pressure, measured across three 24-hour periods per juvenile subject. ON/OFF indicates the light (i.e., active) and dark phases of the daily cycle.

Basal Cardiophysiology

Basal blood pressure was measured in freely moving monkeys while in their family group and home cage using radiotelemetry. Control and ED marmosets demonstrated a significant

elevation of blood pressure during the day compared with night (within-subject factor of phase, main effect: systolic [$F(1,5) = 86.54$, $p < .0001$; diastolic; $F(1,5) = 16.14$, $p < .01$; Figure 1B and 1C]. Early-deprived marmosets demonstrated a significant elevation in systolic blood pressure relative to their CON siblings throughout the day (CON: 91 ± 3; ED: 104 ± 3) and night (CON: 82 ± 3; ED: 95 ± 3) [treatment main effect: $F(1,5) = 12.03$, $p < .02$; Figure 1B]. Average diastolic blood pressure, in particular, the nocturnal diastolic, was consistently higher in ED versus CON marmosets (day, CON: 66 ± 1; ED: 71 ± 3; night, CON: 60 ± 2; ED: 66 ± 2), but this did not reach statistical significance [$F(1,5) = 4.22$, $p < .10$; Figure 1C]. Control and ED marmosets exhibited significantly higher heart rate (beats/min) during the day compared with the night [within-subject factor of phase, main effect: $F(1,5) = 68.81$, $p < .0001$]. Heart rate did not differ between ED and CON subjects [day, CON: 257 ± 11; ED: 248 ± 12; night, CON: 222 ± 14; ED: 210 ± 16; e.g., main effect: $F(1,5) = .21$, $p > .66$]. There was no effect of parentage on any parameter.

Stimulus–Reward Learning and Motivation

The ED subjects (46.7 ± 3.8 trials) performed significantly fewer trials per session across the three discrimination stages compared with CON subjects [53.3 ± 3.4; main effect of treatment: $F(1,10) = 5.5$, $p < .05$], although the significant parentage × treatment interaction [$F(4,10) = 4.14$, $p < .04$] revealed that the ED effect was more pronounced in some family groups than others. Using the total number of errors committed before achieving the learning criterion, the MANOVA of stage × parentage × treatment revealed a significant interaction between stage × treatment [$F(2,20) = 5.19$, $p < .02$], a significant main effect of stage [$F(2,20) = 34.32$, $p < .0001$], and a significant main effect of treatment [$F(1,10) = 9.39$, $p < .02$]. As can be seen in Figure 2A, both ED and CON marmosets committed more errors when performing VR or VRR than VD. A posteriori separate analysis of each discrimination stage revealed that ED marmosets did not make significantly more errors than CON when performing VD [$F(1,10) = .26$, $p > .62$; Figure 2A], indicating that their ability to form novel visual stimulus–reward associations was unaffected by their early-life experience; however, ED marmosets made significantly more reversal errors than CON at VR [$F(1,10) = 6.01$, $p < .04$] and at VRR [$F(1,10) = 18.54$, $p < .002$], indicating that this abnormal postnatal experience chronically impaired their ability to acquire or utilize information (or both) on the altered emotional significance of conditioned stimuli. One possible explanation for this impairment was increased perseveration, that is, impaired ability to inhibit a response to a previously rewarded stimulus; however, when we assessed the extent to which subjects' performed perseverative sessions, defined as a VR or VRR session in which the number of errors (responses to the former S+) was at least as high as the number predicted by the binomial probability distribution at probability .05, then there was no strong evidence that the inferior reversal performance of ED monkeys was a result of increased perseveration (Figure 2B). Rather, ED subjects required several more sessions to progress from chance-level responding, typically focused on one stimulus position (left or right), to preferential responding to the reinforced stimulus under VR and VRR conditions (Figure 2B), which was not the case under VD conditions.

We then assessed the effect of ED on the emotional significance of the innate reward per se, using the PR schedule of reinforcement. Subjects were the same as those that had been tested in VD/VR/VRR and the offspring of two additional groups that could be trained to respond to a single stimulus but not to perform discriminations. In terms of body weight, at the time of PR schedule testing, ED (331 ± 10 g) and CON (346 ± 9 g) marmosets were similar [$F(1,14 = 2.05$, $p > .17$]. The ED marmosets performed significantly less total PR responses than did CON [treatment main effect: $F(1,14) = 6.05$, $p < .03$; Table 2]. As a consequence, ED marmosets obtained significantly less reinforcements than did CON [treatment main effect: $F(1,14) = 5.31$, $p < .04$; Table 2]. The reduced responding of the ED marmosets was in part due to the reduced duration of responding for reinforcement: the interval between the session onset and

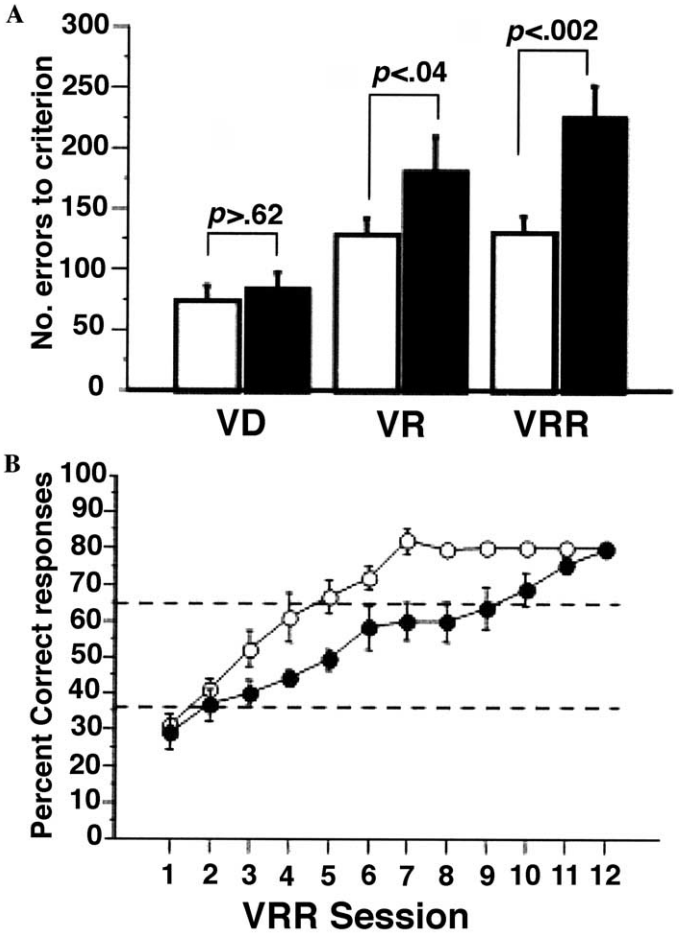


Figure 2. (A) Chronic effects of marmoset early deprivation (ED; filled symbols) on simple visual discrimination (VD) and reversal (VR) relative to control siblings (CON; open symbols). Values are mean ± SEM errors to achieve the learning criterion for each treatment at each stage. The ED marmosets were without significant effects on VD but significantly impaired on VR and re-reversal (VRR) performance. *p* values depict the probabilities obtained in one-way analyses of variance (ANOVAs) performed a posteriori for each stage separately; for this purpose, the mean square error term used was that obtained in the original two-way ANOVA. **(B)** Mean ± SEM percent correct responses per daily session by the ED and CON subjects when tested on VRR learning. Session 1 was the first session conducted with each subject following the animal's achieving the VR learning criterion. According to the binomial probability distribution with $p = q = .5$ and trial $n = 50$, the lower and upper dashed lines depict, respectively, the low and high values for percent (correct) responses to the new S+ that have a chance probability occurrence of .05 in a daily session. Percent correct sessions at or below the lower dashed line were defined as perseverative sessions, and those at or above the upper dashed line were defined as reversal sessions. On VR and VRR learning, ED marmosets were not more perseverative than CON but did perform more sessions than CON at chance level.

Table 2. Effects of Early Deprivation on Motivation for Reward in Adolescent Marmosets

	Total PR Responses	Total PR Reinforcements	Time Elapsed at Last PR Response (sec)
CON	107.5 ± 15.3	12.1 ± 1.2	704.0 ± 55.1
ED	52.8 ± 25.7[a]	8.7 ± 1.6[b]	548.2 ± 79.8[c]

CON, control marmosets; ED, early deprivation marmosets; PR, progressive reinforcement.

Values are M ± SEM ($n = 14$, each group).

[a] $p < .03$ versus CON.

[b] $p < .04$ versus CON.

[c] $p < .02$ versus CON.

final screen-touch response was significantly reduced in ED relative to their CON siblings [treatment main effect: $F(1,14) = 7.2$, $p < .02$; Table 2]. When ED and CON subjects in four of these groups were presented with an excess of freely available reward, they consumed similar volumes: ED, 14.9 ± 2.8 mL; CON, 11.1 ± 1.6 mL [$F(1,8) < 1$, $p > .96$]. Thus there was no evidence that reduced motivation to obtain reward in ED monkeys was related to altered consummatory behavior. Finally, for each of the parameters given in Table 2, there was a significant main effect of parentage [e.g., reinforcements obtained: $F(6,14) = 4.61$, $p < .01$] that was not in interaction with treatment.

Discussion

Our model of parent–infant deprivation has provided strong and novel evidence that early-life stress leads to chronic alteration of the functioning of vital homeostatic and reward systems in a nonhuman primate. As such, it represents a major advance in our potential to understand the importance of the early environment in primates and the mechanisms via which it affects the phenotype in both the short and long term. Human infants who experience physical and emotional deprivation within the primary caregiver relationship also exhibit abnormal physiologic characteristics and are at high long-term risk for affective disorder, which further emphasizes the biomedical importance of models such as that described here.

Norepinephrine is a brain neurotransmitter as well as a neurotransmitter and hormone within the sympathetic branch of the autonomic nervous system (ANS) (Chrousos and Gold 1992). The majority of urinary NE is probably derived from adrenomedullary endocrine activity (Maas and Leckman 1983) and as such provides an indirect but noninvasive marker of basal activity in this sympathetic ANS gland. In CON marmosets the developmental profile and even the absolute levels of urinary NE were similar to those described in humans (Koslow et al 1983; Marchese et al 1997); however, the most striking analogy with the human situation was the marmoset's chronic response to ED in terms of basal NE hyperactivity: following early-life adversity, children with MDD exhibit NE basal hyperactivity (De Bellis et al 1999), and in human adulthood, MDD presents with elevated basal NE levels in urine, blood, and cerebrospinal fluid (Koslow et al 1983; Wong et al 2000). The elevated urinary NE observed in ED marmosets could reflect the chronic consequence of repeated stress in infancy; specifically, it could constitute allostatic traits of high set points for basal activity in the sympathetic ANS (McEwen 2000). Corroborative evidence for this interpretation was provided by using radiotelemetry to monitor cardiophysiologic function in CON and ED juvenile marmosets under undisturbed home-cage conditions and for continuous 24-hour periods (Schnell and Wood 1993). We could demonstrate that ED leads to elevated basal blood pressure in the absence of a significant change in heart rate in the marmoset monkey. It is important to note here that the evidence that ED leads to elevated blood pressure was obtained in different marmosets from those in which it was demonstrated that ED leads to elevated urinary NE output. It has been demonstrated that deprived children exhibit increased systolic blood pressure and heart rate (Perry 1994) and that young women maltreated as children and suffering MDD exhibit increased heart rate stress reactivity (Heim et al 2001). To summarize the chronic effects of ED on stress-sensitive homeostatic systems observed in this prospective primate study, noradrenergic basal hyperactivity and basal systolic hypertension are both consistent with allostasis theory (McEwen 2000) and an altered set point in ANS functioning due to the neglect-like stress experienced in infancy. Taken together with the published evidence for reduced urinary basal GC levels in the same ED marmosets (Dettling et al 2002a, 2002b), it is clear that ED exerts a marked and chronic impact on homeostatic system function analogous, in several respects, to the developmental abnormalities associated with human early-life adversity.

Abnormal functioning of physiologic stress systems is often associated with disruption of neuropsychologic processes as measured using behavioral end points. This has been demonstrated in rodents (Caldji et al 2000; Robbins et al 1996), monkeys (Coplan et al 1996; Kraemer 1992), and humans (Penza et al 2003) and is testimony to the dialogue between brain, physiology, and behavior, including mutual chemical messenger systems (e.g., NE, GC). In our context, the regulation and representation of the emotional significance of environmental stimuli (rewards and punishers) is of marked importance, particularly given the recent multidisciplinary evidence that disturbances of these processes constitute the major symptoms of MDD (Davidson et al 2002; Drevets 2001; Rolls 2000). We recently described some chronic effects of ED on behavioral responses to psychosocial aversive challenge, in the form of separation from the family group in a novel environment (Dettling et al 2002a). In the same subjects as those used in this study, juvenile ED marmosets exhibited less locomotor activity and emitted less social vocalizations than did CON and were in significantly less contact with the father following reunion in the novel environment (Dettling et al 2002a). Although only a subset of these subjects were reported on in this earlier paper, all of the subjects in the study reported here were exposed to the psychosocial aversive challenge.

Building on this earlier evidence for ED leading to increased behavioral responsiveness to aversive challenge, we used conditioned behavior in this study to investigate effects of ED on emotional processing of rewarding environments. In the simple–reversed visual discrimination learning task, ED marmosets did not exhibit a deficit in simple discrimination, indicating ED did not impair simple conditioning to a primary reinforcer. The deficit demonstrated by ED marmosets in visual reversal and re-reversal discrimination indicates a chronic impairment of the ability to acquire or utilize information about changes in the emotional significance of environmental stimuli, an ability important in the changing circumstances that typify primate and human environments—for example, social relationships (Roberts and Wallis 2000; Rolls 2000). A putative explanation of the ED reversal/re-reversal learning deficit would be increased perseveration, which is increased in MDD (Silberman et al 1983), but according to our working definition, this was not the case; both ED and CON marmosets performed 1–2 perseverative sessions.

The ED marmosets then performed a higher number of chance-level sessions than CON. During such sessions, most ED and CON subjects exhibited a bias to respond to one stimulus position (left or right) but not to S+ or S−. In contrast, whereas CON subjects exhibited an average of two such sessions and then began to "follow," that is, approach and respond to, the new S+ regardless of the side it was presented, ED subjects required an average of seven sessions before they began to respond preferentially to the new S+. Although with training marmosets become proficient at reversal/re-reversal learning (Pearce et al 1998; Spinelli et al 2004), it is possible that the initial response of marmosets naïve to VR/VRR is that S's, and therefore S-related behavior, have become irrelevant to obtaining reinforcement. If this is the case, then prolonged periods of chance-level VR/VRR performance, as demonstrated by ED subjects, could reflect increased vulnerability to a perceived loss of environmental control, at least with respect to obtaining positive reinforcers. Appetitive tasks designed specifically to assess this, informed by general learned irrelevance theory (Linden et al 1997), are currently being developed. The cognitions of depressed children and adolescents are dominated by sadness related to personal inadequacy or inability (Meerum Terwogt and Stegge 2001), which highlights the potential construct validity of this primate environmental model. In adult marmosets, impaired visual reversal discrimination was induced by excitotoxic lesioning of the orbitofrontal (prefrontal) cortex (Dias et al 1997), a neocortical region revealed by magnetic resonance imaging to be hypoactive in MDD (Drevets 2001). In contrast to our study, however, the deficit induced by orbitofrontal lesioning was primarily one of increased perseveration (Dias et al 1997). Finally, it is noteworthy that ED subjects made more errors on re-reversals than reversals, possibly reflecting an increased sensitivity to proactive interference compared with CON; by the VRR stage both stimuli had been rewarded, and the potential load of proactive interference would have been high (Mackintosh 1974).

Under the conditions of the VD/VR/VRR task, random responding to S+ and S− will yield reinforcement of every second response on average, such that the motivational demands of the task are low. Nonetheless, ED led to a small but consistent reduction in the number of daily trials performed per session relative to CON, possibly related to lower interest in reward, but also possibly to the reduced rate of reinforcement under VR/VRR conditions. Using the PR schedule of reinforcement, clear evidence was obtained that the reward constituted less of an incentive to ED marmosets than it did to CON. The reduction in appetitive behavior to obtain reward was not accompanied by reduced consummatory behavior under conditions in which the reward was presented to the subjects directly; indeed, both CON and ED subjects consumed considerably more reward than they obtained during PR schedule testing. Therefore, the stringent conditions of the PR schedule revealed that ED leads to development of a trait of reduced incentive value of a primary reinforcer, suggesting mild anhedonia. The diminished interest in reward of ED subjects as revealed in the PR schedule is unlikely to have been a consequence of their experiencing more nonrewarded trials in the preceding VD/VR/VRR task. We draw this conclusion on the basis of the well-documented partial reinforcement extinction effect, that is, reduction in the probability of reinforcement increases subsequent resistance of the operant response (which was the same in both the VD/VR/VRR and PR schedule tasks) to extinction (Mackintosh 1974). The combination of VD/VR/VRR and PR schedule testing revealed that ED leads to chronic effects on environmental processing, including evidence for traits of increased vulnerability to (perceived) loss of environmental control with respect to positive reinforcement and of decreased interest in such reinforcement. Future studies will need to explore these psychologic changes further to validate their relevance to depression, and in this respect the investigation of altered processing of contrast effects, that is, responses to changes in reward intensity (e.g., Matthews et al 1996; Matthews and Robbins 2003), will be important. It is now also important to investigate the neurobiological correlates of the consistent physiologic and behavioral effects of ED identified in these marmoset monkeys, informed by the human evidence for both the chronic neurodevelopmental effects of early-life adversity (Caspi et al 2003; Penza et al 2003; Teicher et al 2003) and the neuropathology of affective disorders (Drevets 2001; Harrison 2002).

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American Psychiatric Association (1994): *Diagnostic and Statistical Manual of Mental Disorders,* 4th ed. Washington, DC: Author.

Bennett AJ, Lesch KP, Heils A, Long JC, Lorenz JG, Shoaf SE, et al (2002): Early experience and serotonin transporter gene variation interact to influence primate CNS function. *Mol Psychiatry* 7:118–122.

Blumberg MS, Sokoloff G (1998): Thermoregulatory competence and behavioral expression in the young of altricial species-revisited. *Dev Psychobiol* 33:107–123.

Bowlby J (1962): *Attachment and Loss.* New York: Basic Books.

Caldji C, Francis D, Sharma S, Plotsky PM, Meaney MJ (2000): The effects of early rearing environment on the development of GABAA and central benzodiazepine receptor levels and novelty-induced fearfulness in the rat. *Neuropsychopharmacology* 22:219–229.

Caspi A, Sugden K, Moffitt TE, Taylor A, Craig IW, Harrington H, et al (2003): Influence of life stress on depression: Moderation by a polymorphism in the 5-HTT gene. *Science* 301:386–389.

Champoux M, Bennett AJ, Shannon C, Higley JD, Lesch KP, Suomi SJ (2002): Serotonin transporter gene polymorphism, differential early rearing, and behavior in rhesus monkey neonates. *Mol Psychiatry* 7:1058–1063.

Chrousos GP, Gold PW (1992): The concepts of stress and stress system disorders: Overview of physical and behavioral homeostasis. *JAMA* 9:1244–1252.

Coplan JD, Andrews MW, Rosenblum LA, Owens MJ, Friedman S, Gorman JM, Nemeroff CB (1996): Persistent elevations of cerebrospinal fluid concentrations of corticotropin-releasing factor in adult nonhuman primates exposed to early-life stressors: Implications for the pathophysiology of mood and anxiety disorders. *Proc Natl Acad Sci U S A* 93:1619–1623.

Davidson RJ, Pizzagalli D, Nitschke JB, Putnam K (2002): Depression: Perspectives from affective neuroscience. *Ann Rev Psychol* 53:545–574.

De Bellis MD, Baum AS, Birmaher B, Keshavan MS, Eccard CH, Boring AM, et al (1999): A.E. Bennett Research Award. Developmental traumatology. Part I: Biological stress systems. *Biol Psychiatry* 45:1259–1270.

Dettling AC, Feldon J, Pryce CR (2002a): Early deprivation and behavioural and physiological responses to social separation/novelty in the marmoset. *Pharm Biochem Behav* 73:259–269.

Dettling AC, Feldon J, Pryce CR (2002b): Repeated parental deprivation in the infant common marmoset (*Callithrix jacchus*, primates) and analysis of its effects on early development. *Biol Psychiatry* 52:1037–1046.

Dias R, Robbins TW, Roberts AC (1997): Dissociable forms of inhibitory control within prefrontal cortex with an analog of the Wisconsin card sort test: Restriction to novel situations and independence from on-line processing. *J Neurosci* 17:9285–9297.

Drevets WC (2001): Neuroimaging and neuropathological studies of depression: Implications for the cognitive-emotional features of mood disorders. *Curr Opin Neurobiol* 11:240–249.
Harlow HF (1958): The nature of love. *Am Psychol* 13:673–685.
Harlow HF, Harlow MK (1962): Social deprivation in monkeys. *Sci Am* 207:136–139.
Harrison PJ (2002): The neuropathology of primary mood disorder. *Brain* 125:1–22.
Heim C, Nemeroff CB (2001): The role of childhood trauma in the neurobiology of mood and anxiety disorders: Preclinical and clinical studies. *Biol Psychiatry* 49:1023–1039.
Heim C, Newport DJ, Bonsall R, Miller AH, Nemeroff CB (2001): Altered pituitary-adrenal axis responses to provocative challenge test in adult survivors of childhood abuse. *Am J Psychiatry* 158:575–581.
Hinde RA, Spencer-Booth Y, Bruce M (1966): Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature* 210:1021–1023.
Hofer MA (1994a): Early relationships as regulators of infant physiology and behavior. *Acta Paediatr Suppl* 397:9–18.
Hofer MA (1994b): Hidden regulators in attachment, separation, and loss. *Monogr Soc Res Child Dev* 59:192–207 250–283.
Huot RL, Thrivikraman KV, Meaney MJ, Plotsky PM (2001): Development of adult ethanol preference and anxiety as a consequence of neonatal maternal separation in Long Evans rats and reversal with antidepressant treatment. *Psychopharmacol* 158:366–373.
Kendler KS, Gardner CO, Prescott CA (2002): Toward a comprehensive developmental model for major depression in women. *Am J Psychiat* 159:1133–1145.
Koslow SH, Maas JW, Bowden CL, Davis JM, Hanin I, Javaid J (1983): CSF and urinary biogenic amines and metabolites in depression and mania. *Arch Gen Psychiatry* 40:999–1010.
Kraemer GW (1992): A psychobiological theory of attachment. *Behav Brain Sci* 15:493–541.
Kraemer GW, Ebert MH, Schmidt DE, McKinney WT (1989): A longitudinal study of the effect of different social rearing conditions on cerebospinal fluid norepinephrine and biogenic amine metabolites in rhesus monkeys. *Neuropsychopharmacol* 2:175–189.
Ladd CO, Huot RL, Thrivikraman KV, Nemeroff CB, Meaney MJ, Plotsky PM (2000): Long-term behavioral and neuroendocrine adaptations to adverse early experience. *Prog Brain Res* 122:81–103.
Levine S (1994): The ontogeny of the hypothalamic-pituitary-adrenal axis: The influence of maternal factors. *Ann N Y Acad Sci* :275–293.
Linden DR, Savage LM, Overmier JB (1997): General learned irrelevance: A Pavlovian analog to learned helplessness. *Learn Motiv* 28:230–247.
Liu D, Caldji C, Sharma S, Plotsky PM, Meaney MJ (2000): Influence of neonatal rearing conditions on stress-induced adrenocorticotropin responses and norepinephrine release in the hypothalamic paraventricular nucleus. *J Neuroendocrinol* 12:5–12.
Maas JW, Leckman JF (1983): Relationships between central nervous system noradrenergic function and plasma and urinary MHPG and other norepinephrine metabolites. In: Maas JW, editor. *MHPG: Basic Mechanisms and Psychopathology*. New York: Academic Press, 33–43.
Mackintosh NJ (1974): *The Psychology of Animal Learning*. London: Academic Press.
Marchese N, Canini S, Fabi L, Famularo L (1997): Paediatric reference values for urinary catecholamine metabolites evaluated by high performance liquid chromatography and electrochemical detection. *Eur J Clin Chem Clin Biochem* 35:533–537.
Matthews K, Robbins TW (2003): Early experience as a determinant of adult behavioural responses to reward: The effects of repeated maternal separation in the rat. *Neurosci Biobehav Rev* 27:45–55.
Matthews K, Wilkinson LS, Robbins TW (1996): Repeated maternal separation of preweanling rats attenuates behavioral responses to primary and conditioned incentives in adulthood. *Physiol Behav* 59:99–107.
McCormick CM, Kehoe P, Kovacs S (1998): Corticosterone release in response to repeated, short episodes of neonatal isolation: Evidence of sensitization. *Int J Dev Neurosci* 16:175–185.
McEwen BS (2000): The neurobiology of stress: From serendipity to clinical relevance. *Brain Res* 886:172–189.
Meerum Terwogt M, Stegge H (2001): The development of emotional intelligence. In: Goodyer IM, editor. *The Depressed Child and Adolescent*. Cambridge: Cambridge University Press, 24–45.
O'Connor TG, Marvin RS, Rutter M, Olrick JT, Britner PA, and the English and Romanian Adoptees Study Team (2003): Child-parent attachment following early institutional deprivation. *Dev Psychopathol* 15:19–38.
Paul IA, English JA, Halaris A (2000): Sucrose and quinine intake by maternally-deprived and control rhesus monkeys. *Behav Brain Res* 112:127–134.
Pearce PC, Crofts HS, Muggleton NG, Scott EAM (1998): Concurrent monitoring of EEG and performance in the common marmoset: A methodological approach. *Physiol Behav* 63:591–599.
Perry BD (1994): Neurobiological sequelae of childhood trauma: PTSD in children. In: Murburg M, editor. *Catecholamine Function in Posttraumatic Stress Disorder: Emerging concepts*. Washington DC: American Psychiatric Press, 173–189.
Penza KM, Heim C, Nemeroff CB (2003): Neurobiological effects of childhood abuse: Implications for the pathophysiology of depression and anxiety. *Arch Women Ment Health* 6:15–22.
Plotsky PM, Meaney MJ (1993): Early, postnatal experience alters hypothalamic corticotropin-releasing factor (CRF) mRNA, median eminence CRF content and stress-induced release in adult rats. *Mol Brain Res* 18:195–200.
Pryce CR (1996): Socialization, hormones and the regulation of maternal behavior in nonhuman simian primates. *Adv Stud Anim Behav* 25:423–473.
Pryce CR, Feldon J (2003): Long-term neurobehavioural impact of the postnatal environment in rats: Manipulations, effects and mediating mechanisms. *Neurosci Biobehav Rev* 27:57–71.
Robbins TW, Jones GH, Wilkinson LS (1996): Behavoural and neurochemical effects of early social deprivation in the rat. *J Psychopharmacol* 10:39–47.
Roberts AC, Robbins TW, Everitt BJ (1988): The effects of intradimensional and extradimensional shifts on visual discrimination learning in humans and non-human primates. *Q J Exp Psychol B* 40:321–341.
Roberts AC, Wallis JD (2000): Inhibitory control and affective processing in the prefrontal cortex: Neuropsychological studies in the marmoset. *Cereb Cortex* 10:252–262.
Rolls ET (2000): Precis of *The brain and emotion*. *Behav Brain Sci* 23:177–234.
Rutter M (2003): The interplay of nature, nurture, and developmental influences: The challenge ahead for mental health. *Arch Gen Psychiatry* 59:996–1000.
Schnell CR, Wood JM (1993): Measurement of blood pressure and heart rate by telemetry in conscious, unrestrained marmosets. *Am J Physiol* 264:H1509–516.
Seay BM, Hansen EW, Harlow HF (1962): Mother-infant separation in monkeys. *J Child Psychol Psychiatry* 3:123–132.
Shalev U, Kafkafi N (2002): Repeated maternal separation does not alter sucrose-reinforced and open-field behaviors. *Pharm Biochem Behav* 73:115–122.
Silberman EK, Weingartner H, Post RM (1983): Thinking disorders in depression. *Arch Gen Psychiatry* 40:775–780.
Spencer-Booth Y, Hinde RA (1971a): Effects of 6 days of separation from mother on 18- to 32-week old rhesus monkeys. *Anim Behav* 19:174–191.
Spencer-Booth Y, Hinde RA (1971b): Effects of brief separations from mothers during infancy on behaviour of rhesus monkeys 6–24 months later. *J Child Psychol Psychiat* 12:157–172.
Spinelli S, Pennanen L, Dettling AC, Feldon J, Higgins GA, Pryce CR (2004): Performance of the marmoset monkey on computerized tasks of attention and working memory. *Brain Res Cogn Brain Res* 19:123–137.
Spitz RA, Wolf KA (1946): Anaclitic depression: An inquiry into the genesis of psychiatric conditions in early childhood. *Psychoanal Study Child* 2:313–342.
Sutherland SN, Mackintosh NJ (1971): *Mechanisms of Animal Discrimination Learning*. New York: Academic Press.
Teicher MH, Andersen SL, Polcari A, Anderson CM, Navalta CP, Kim DM (2003): The neurobiological consequences of early stress and childhood maltreatment. *Neurosci Biobehav Rev* 27:33–44.
Wong M-L, Kling MA, Munson PJ, Listwak S, Licinio J, Prolo P, et al (2000): Pronounced and sustained central hypernoradrenergic function in major depression with melancholoic features: Relation to hypercortisolism and corticotropin-releasing hormone. *Proc Natl Acad Sci U S A* 97:325–330.
Zhang L-X, Levine S, Dent GW, Zhan Y, Zing G, Okimoto DK, et al (2002): Maternal deprivation increases cell death in the infant rat brain. *Dev Brain Res* 133:1–11.